

Fluid Mechanics Project

Data Driven Analysis of Fluid Flows

Lavish Kumar(B22ME035)
Himanshu Chauhan(B22ME025)

November 16, 2025

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1 Machine Learning in Fluid Mechanics

1.1 Introduction

As simple aerodynamic configurations under normal conditions exhibit complex flows having wide range of temporal and spatial features. So, to analyse these flows, we have to extract physically important features, or modes in first step of analysis. This step starts with a modal decomposition of an experimental or, numerical dataset of the flowfield, or of an operator relevant to the system. So, here we will describe some of the dominant techniques for accomplishing these modal decompositions and analyses that have seen a surge of activity in recent decades. Here we will discuss some modal analysis techniques like Eigenvalue Decomposition, Singular Value Decomposition (SVD), Proper Orthogonal Decomposition (POD).

1.2 Eigenvalue Decomposition

Eigenvalue decomposition is performed on a square matrix. The eigenvalues and eigenvectors of a matrix (linear operator) capture the directions in which vectors can grow or shrink. For a given matrix $A \in \mathbb{C}^{n \times n}$, a vector $v \in \mathbb{C}^n$ and a scalar $\lambda \in \mathbb{C}$ are called an eigenvector and an eigenvalue, respectively, of A if they satisfy

$$Av = \lambda v$$

Eigenvectors are unique up to a complex scalar. The eigenvectors obtained from computer programs are commonly normalized such that they have unit magnitude. The set of all eigenvalues of A is called a spectrum of A.

1.3 Singular Value Decomposition

Singular value decomposition (SVD) is one of the most important matrix factorizations, generalizing the eigendecomposition to rectangular matrices. SVD has many uses and interpretations, especially for dimensionality reduction, where it is possible to use SVD to obtain optimal low-rank matrix approximations.

1.4 Proper Orthogonal Decomposition

Proper orthogonal decomposition is a modal decomposition technique that extracts modes based on optimizing the mean square of the field variable which is being examined. It provides an objective algorithm to decompose a set of data into a minimal number of basis functions or modes to capture as much energy as possible. The method itself is known under a variety of names in different fields: POD, principal component analysis (PCA), empirical eigenfunction decomposition, and others. In applications of POD to a fluid flow, we start with a vector field $q(\xi, t)$ (e.g., velocity) with its temporal mean $\bar{q}(\xi)$ subtracted and assume that the unsteady component of the vector field can be decomposed in the following manner:

$$q(\xi, t) - \bar{q}(\xi) = \sum_j a_j \phi_j(\xi, t) \quad (1)$$

where $\phi_j(\xi, t)$ and a_j represent the modes and expansion coefficients, respectively. Here, ξ denotes the spatial vector.

This expression represents the flow field in terms of a generalized Fourier series for some set of basis functions $\phi_j(\xi, t)$. In the framework of the POD, we seek the optimal set of basis functions for given flow field data.

POD is one of the most widely used techniques in analyzing fluid flows. There are a large number of variations of POD technique, with applications including fundamental analysis of fluids flows, reduced-order modeling, data compression/reconstruction, flow control, and aerodynamic design optimization.

2 New Ideas

2.1 Modeling the near-wall region of wall-bounded turbulence

Nearly 20 years ago, a method was developed to predict the relevant features of turbulent channel flow close to the wall by using deep neural networks. In addition to the interest of this work from the perspective of modeling turbulence, and potentially developing wall models, they also established connections between neural networks and traditional methods such as proper-orthogonal decomposition (POD). In particular, they showed that by restricting the neural network model to be linear, the network essentially learns the

features produced by standard POD. This topic has received renewed attention in recent years due to the possibility of training deeper neural network models with larger data sets, as well as the emergence of novel learning architectures .

2.2 Development of inflow conditions for turbulence simulations

This is a critical area for CFD when it comes to achieving high-Reynolds-number (Re) conditions and simulating complex geometries. Spatially developing turbulent boundary layers (TBLs) require very long domains to reach high Reynolds numbers, and if part of the low-Re region can be replaced by an adequate inflow condition, simulations at the relevant high-Re regime can become feasible. Similarly, if a simulation is designed to study the turbulent flow around a complex array of obstacles, it is beneficial to replace the inflow section of the simulation by a suitable inflow condition, thus yielding significant computational savings. Traditional approaches to these inflows have relied on various filters, recycling approaches, and imposing synthetic turbulent structures. However, recent progress in ML has enabled the use of deep-learning approaches to produce robust inflow conditions, including the use of modern architectures, such as transformers.

2.3 Development of boundary conditions for external-flow simulations

One of the challenges of simulating external flows, e.g. the flow past an airfoil, is the need for large computational domains to adequately reproduce the far-field conditions and the pressure distribution around the wing. While the resolution of the portion of the domain near the far field is often already quite coarse, being able to replace this region entirely by a suitable boundary condition would lead to significant computational savings. Furthermore, it is important to be able to establish prescribed pressure-gradient (PG) conditions so as to study PG TBLs and their non-trivial flow-history effects. To this end, a number of ML approaches have been recently developed for effective boundary-condition design. In particular, Gaussian-process regression has been successfully employed to optimize the shape of the domain to prescribe a particular PG distribution.

2.4 Novel approaches for flow control

Machine learning (ML) is helping to develop new ways of controlling fluid flow. In the past, scientists relied on simulations to create effective control methods, which were then tested through experiments. One common approach is called active flow control, which uses energy to manipulate turbulent flows, aiming to reduce drag or improve mixing. The most efficient method is closed-loop control, where measurements of the flow's current state guide the control actions in real-time. Opposition control is a traditional closed-loop method that uses a jet to counteract wall disturbances.

Now, ML techniques offer more advanced control strategies. Deep reinforcement learning discovers new control methods across various flow scenarios, including turbulent flow control. These ML-based approaches are more sophisticated than the traditional simulation-based methods, and they hold promise for further improving fluid flow control.

3 Proper Orthogonal Decomposition

Here we had used Principal Component Analysis(PCA) instead of POD . Implementation of both PCA and POD are almost similar.

3.1 Mathematical Formulations Used

Compute Mean Image : Calculate the mean image by taking the mean across all images in the dataset. This is done by averaging pixel values across corresponding pixels of all images.

Compute Covariance Matrix : Subtract the mean image from each image to center the data. Reshape the images into vectors. Compute the covariance matrix, which represents the relationships between the different pixel values across images.

Power Iteration : This function finds the dominant eigenvector (principal component) of a matrix using the power iteration method. We start with a random vector. Iterate several times, multiplying the matrix by the vector and normalizing the result. The resulting vector converges to the dominant eigenvector of the matrix. We also calculate the corresponding eigenvalue.

PCA : We compute the mean image and covariance matrix using the previously defined functions. Iterate to find the principal components. Subtract the contribution of each principal component from the covariance matrix.

Return the mean image and the principal components (modes) as the output of PCA.

Reconstruct Images : Firstly, reshape the images into vectors. Then center the images by subtracting the mean image. Project the centered images onto the principal components to obtain coefficients. Reconstruct the images using the coefficients and principal components, and add back the mean image. Reshape the reconstructed vectors into images.

3.2 Significance of POD Modes

Dimensionality Reduction : Principal Orthogonal Decomposition (POD) simplifies high-dimensional data. Its key purpose is to reduce dimensionality. It captures major variations through modes. POD represents data concisely. This increases computational efficiency. It also enhances data understanding, a significant benefit.

Characterization of System Dynamics : POD modes display the dominant spatial or temporal patterns in the dataset. For fluid dynamics, the modes might show coherent flow structures or turbulence. In structural mechanics, they could match deformation modes or vibration patterns. Analyzing these modes helps researchers understand the system's dynamics and behavior.

Model Reduction and Reconstruction : POD modes are helpful for creating simpler models of complex systems. By keeping only the most important modes, these simpler models can show the main dynamics while using way less computing power. Also, POD modes can reconstruct the original data with very little information loss, making tasks like denoising or super-resolution possible.

4 Effect of Noise on POD Modes

4.1 How does the noise affect different modes?

Noise in images can introduce additional variability, which can impact the calculation of modes during Proper Orthogonal Decomposition (POD). High-frequency noise often dominates higher modes, causing distortion in their

spatial patterns. Lower modes may still capture the main features of the underlying signal, but they could also be affected, especially if the noise level is substantial.

4.2 How does the energy transcend through modes?

The energy distribution among modes shows how important each mode is for representing the variability in the data. Higher modes usually capture finer details or noise. Lower modes, on the other hand, represent the dominant patterns or structures. As we move from lower to higher modes, the energy tends to decrease. This means that a smaller number of modes can explain most of the variability in the data.

4.3 Can you give any statistical evidence of the energy changes in the modes?

We can analyze the variance or energy associated with each mode to understand their significance. Computing the cumulative energy distribution allows us to determine how much of the total energy is captured by a certain number of modes. Additionally, statistical measures such as mean, median, and standard deviation of mode energies can provide insights into their distribution and variability.

4.4 Do different noise types have different type of response in terms of how the energy of the modes changes?

Several noise varieties impact mode energies distinctly. For instance, Gaussian noise chiefly influences higher-frequency elements, resulting in amplified energy levels in higher modes. Conversely, salt-and-pepper noise generates outliers, affecting the statistical properties of the data and potentially influencing multiple modes.

4.5 How does the magnitude of noise affect the modal energy?

Noise, when increased, often boosts the energy in higher modes. However, it might lower the energy in lower modes. This happens because more intense noise creates greater variety in the data. The higher modes capture this variability. Dominant patterns, which lower modes represent, may stay relatively steady with moderate noise levels. Yet, substantial noise can impact these lower modes.

4.6 What sort of noise is best suited to be used for POD for artificial analysis?

The choice of noise depends on the specific characteristics of the data and the analysis goals. Gaussian noise is often used as it models natural variations and is well-suited for analyzing linear systems. However, if the data exhibits non-linear behavior or contains impulsive noise, other types of noise such as salt-and-pepper noise or speckle noise may be more appropriate.

5 Super Resolving/Denoising

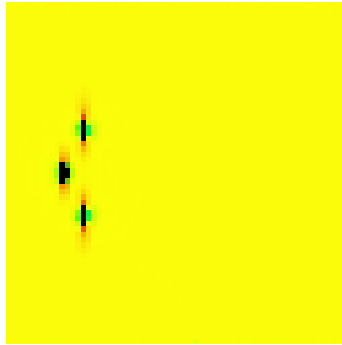


Figure 1: Original Image

Here it clearly seems that in denoised image there is reduction of noise and there is uniform color in denoised image.

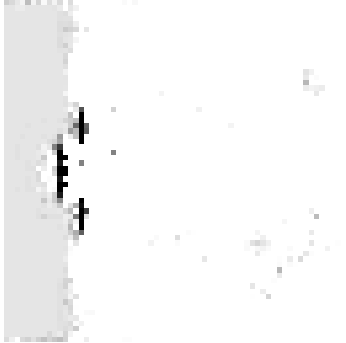


Figure 2: PCA Image

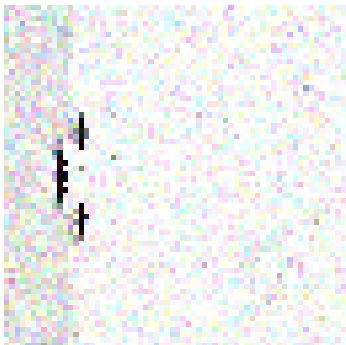


Figure 3: Noisy Image

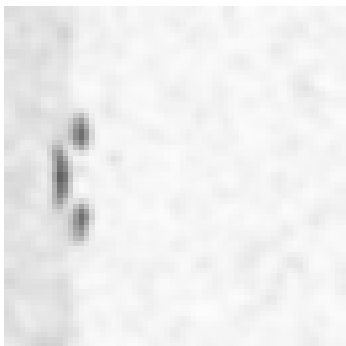


Figure 4: Denoised Image